1. Outstanding work

(current work is in `opencode -s ses_ef6ed9507ffe4pRp5Yo4scC07k`) Only unresolved tasks are listed here, in ascending order. Numbers are permanent; append new tasks starting at 339. The historical checklist and detailed work diary are preserved in the archived todo file. See the completion summary for the latest work.

26. [X] (Deferred) once in a while the text everywhere breaks (see pic) some sort of leak?

114. ☐ crash recovery: restore unsaved buffers and all document windows after an unexpected exit

116. ☐ workspace search and replace: preview replacements across files, support regex capture groups, and provide undo

117. ☐ large-project performance: share repository status between windows and measure indexing, preview, and typing latency (ongoing optimization, using the profiling foundation in item 2)

163. [x] Allow for the pdf preview to pop out into a different window. if I close it, the preview should go back to the window. I think we can essentially reuse the main window code for the extra window, just dont let it open a bottom panel, or explorer panel, or code panel. and the two windows should be linked in their actions - if in the main window we toggle the preview back on whether in split or just preview mode, then this extra preview window should be closed. and, if we close the extra preview window, it should open back in the main window as either preview or split mode.

182. ☐ pretty animation for dragging tabs

223. ☐ prove out typst-compatible binaries like calepin

234. [X] Isolate the transient oddly shaped large-window flash reported when

opening Settings with multiple main windows. Duplicate Settings ownership is fixed in 231, but steady-state native observations/framebuffers cannot certify that a short-lived flash is gone. Keep this separate from singleton acceptance.

237. ☐ Consolidate Problems, index and preview navigation/focus handoff after

reproducing the native failure. Test document/range routing and secondary owners; do not equate helper tests with verified native word-navigation behavior. Implementation: `src/app/navigation.rs` owns destination conversion, diagnostic targeting, current-file/deferred routing and source-range handoff. File links and Explorer entries share routing; Problems retain already-mapped editor coordinates, while canonical file/LSP positions still use the miTeX mapping. Existing document workflow epoch checks and native-parent → viewport → TextEdit focus order remain. Reproduced a concrete focus failure before fixing it: with Find open, a source jump moved the caret but left Find focused. The existing pending selection now distinguishes Search (retain focus) from Focus (source/editor navigation), rather than introducing a separate focus flag. Find stays open after navigation. Regression coverage exercises all three entry points in root and secondary viewports, Option/Cmd Left/Right, owner-only one-shot focus and Find focus retention. This is deterministic UI coverage, not an observed macOS WebView first-responder handoff; leave the item open for that native acceptance check. Accounting versus the preceding leaf-extraction commit: app.rs 12,840 → 12,647 (−193); navigation module 226 lines; other production adapters +3, for +36 total non-test lines. Tests +181. The added selection intent fixes a demonstrated correctness gap; it is not claimed as a net-negative consolidation. No workers, repaint loops, per-frame source copies or native geometry changes are introduced; the extra branch runs only for a pending selection. No timing speedup is claimed. Verification: failing-before/passing-after Find-open focus regression, all 13 navigation-filter tests, Find focus-retention test, 1,061 full-suite test executions (16 opt-in tests ignored), 13 xtask tests, formatting and strict all-target Clippy pass. Native WebView delivery remains unverified; subsequent steps are tracked below.

251. ☐ (Deferred after 246′s review; rationale in the work diary.)

Consolidate one structured-edit path, starting with formatting result application. Trace its existing document-key and canonical/display checks; share an existing validated edit primitive only where semantics match, retaining format-specific admission. Do not turn completion transactions into a universal framework. Acceptance: stale/reopened-document and out-of-order replies are rejected, Unicode and miTeX ranges remain correct, accepted edits have one undo step, and ordinary Typst incurs no projection or extra source copy. Delete the replaced mutation path. Depends on 246; expand to other edit kinds only in separately scoped follow-ups justified by actual duplication.

253. ☐ (Deferred after 246′s review; rationale in the work diary.)

Narrow native-preview resource ownership for one child-view kind. Use 246′s writer map to select the existing view handle, applied-property cache and teardown paths that must change together. Make their lifecycle operations explicit and delete bypass setters; leave readiness/retry policy in the existing PreviewController. Acceptance: hide/show, replacement, close and repeated teardown release once, unchanged properties cause no native setters, and stale owners cannot reposition a replacement view. Preserve UI-thread ownership and verify native geometry/composition when affected. Do not bundle every preview flag into a second controller. Depends on 246 and relevant native gates 234/237.

264. ☐ Tinymist preview: smooth zoom on large real documents. Continuous pointer-anchored Ctrl-scroll is implemented, but full-SVG layout remains slow. Native preview shortcuts now dispatch to the viewer and mixed inputs accumulate in order. Evaluate a cached viewport bitmap for gesture transitions: capture after rendering settles, not on the first gesture event; bound pixel memory, preserve the cursor anchor, and swap only when sharp content is ready.

265. ☐ Tinymist preview: smooth divider resizing on large real documents. Redundant rerenders are removed; the full SVG still costs about 200 ms to rescale in WKWebView. Validate bounded visible-region rendering and scroll refill before closing these items; the small fixture was insufficient.

266. ☐ vim mode

268. ☐ while adjusting the zoom, we show the user a blank page while it loads. it should always show us the lower res page until it loads the higher res page

283. ☐ connect to TPIX https://typstify.com/tpix

289. ☐ Document and expose a deliberate partial-rendering buffer policy for the Tinymist preview. The pinned frontend currently offers only the boolean `--partial-rendering` switch: its SVG window is derived from the preview scroll position and viewport, expanded to page boundaries, and is not tied to the editor cursor. Decide whether a bounded page/viewport prefetch is worth implementing upstream or in the adapter, then measure scroll refill latency and memory before changing the production default.

290. ☐ (deferred) Add a bounded macOS bitmap transition for the live Tinymist viewport. Keep one in-memory WKWebView snapshot, show it only during an active zoom/divider transition, and replace it after the live preview settles. Capture no image every frame, invalidate stale callbacks when the webview or document changes, preserve pointer anchoring, and verify that the overlay cannot steal editor/preview input. Add deterministic frame/state tests and a native geometry trace before enabling it by default.

295. [x] Removed the legacy raster PDF viewer and PDF.js, including their settings, adapters, assets, render workers and capture surrogate. PDFium owns PDF viewing and framebuffer captures; image viewing and hover thumbnails remain separate.

2. NEW TODOS

1. [x] sometimes tinymist can redirect you to a source file outside the current workspace root (when it sends you to a package file). When this happens, the workspace root switches to the workspace of that file, and that file is opened in a new tab. I agree that it should open in a new tab but not that the workspace root changes. if we nav back to the old tab the workspace root changes back. It should instead, at the top of the code editor, have a small banner that says "You are currently editing a file outside the current workspace root. The workspace root is nx ow [workspace root path]." and have a button that says "Switch to this workspace root" which when clicked, switches the workspace root to the new one. This way, the user is aware of the change and can choose to switch or not.
2. [x] the stage all and commit button / commit all staged button in the git subpanel should be text
3. [x] we need to implement comfy mode - if on, the bg and text of a pdf / tinymist preview should be adjusted to match the theme settings.
4. [x] improve harper typst/tex integration - it needs to ignore string keys of functions and spellchecking on names somehow
5. [x] allow using a tex dist (pdfLaTeX, XeLaTeX, LuaLaTeX) and autodetect installed tex distributions (mactex)
6. [x] implement synctex for tex docs
7. [x] dragging a file from the explorer to the code panel should put the path at the cursor
8. [x] i noticed that the tinymist preview has a bunch of shortcuts. for instance pressing t inverts the view! we should use this instead of repainting the preview from scratch when we toggle dark mode. Make it work smoothly with comfy mode above
9. [x] the above shortcuts are interfering with the find feature of our litle popup. also I found that cmd+A selects all in the code panel - the find bar should be the one that receives thte cmd+A.
10. [x] the TOC doesn't jump to the sections for tinymist
11. [x] the design for the minimised icon is different for both variants (pdf and tinymist preview). And anyway it uses way too much space. fix and make the button more compact
12. [x] the UI of the interface is completely inconsistent in all states - minimised (see above and below todos), unminimised, and with TOC expanded. The UI layout should be shared code, just the wiring can be different
13. [x] instead of a floating minimized button, in fact it should be a button left of the find button in the top. So remove the floating button. Keep the floating window when expanded. And this button should be greyed out inactive if editor is nto in preview or split mode.
14. [x] the floating window should be more thoughtfully designed. the buttons needs to be correctly grouped while not taking too much space.
15. [x] i modified old todo 163 above - this action should be in our floating window
16. [x] tab -> 2 or 4 or x spaces or tab should be settable in settings (default to 2)
17. [x] i noticed in a cjk keyboard typing 。 gave me a . - we should be able to toggle this behavior
18. [x] tex diagnostics/fmt override in mitex
19. [x] detexify / detypstify (https://github.com/kirel/detexify-next / https://detypify.quarticcat.com/) we should just pull the model params/hyperparams out and use rust as much as possible (i hear good things about ax). Add it as an explorer subpanel we can draw in
20. [x] more customizable snippets
21. [x] multiple cursor snippeting: e.g. if we tab autocomplete to `\begin` and then type e.g. `enumerate` the generated `\end` should also be populated with `enumerate`.
22. [x] sometimes, when switching from tex to typ, i get that tinymist is stale, tectonic and harper stays around from the tex file, and i dont get new diagnostics until first keyboard press. after, harper refreshed (yay) but tectonic stuck around (boo)
23. [x] dragging the pdf preview options popup around, it moves in a very janky jumpy way, and the end result is only 50% of the way between start and final mouse position. Also i can only drag from the title "Preview". And the "v" button to close the popup should be a "-" minimise button.

24. ☐ cmd+F when the preview panel is focused should open the preview's extra options panel and focus the find box.
25. ☐ box dragging polish: the preview options panel doesnt move perfectly when dragging and held: sometimes, it keeps the direction previously dragged in with a constant velocity and then stops when box's boundary touches the stationary mouse cursor.
26. ☐ we should remove all top/down/right padding for the terminal windows.
27. ☐ Similarly to how the preview panel has an outline when focused, we need the same for the problems panel
28. ☐ the outline for the code panel should show through the sticky rows, so that we can see it when hovered/focused. You can consider either adjusting the size by a few px on left,right,and top, or just make the borders of the left,right and top of the sticky rows mimic the outline of the code panel.
29. ☐ The code panel has line highlights but it only goes to the length of the longest line. It should always go for the full length of the code panel.
30. ☐ minor polish - converting to inline/display math should by default be automatically followed by a fmt
31. ☐ the table editor needs to allow for adding hlines and whatnot, the colors and border options etc should result in visible changes to the UI, e.g. the cells should have their border colored if the border color is set. Whole table settings and selected cells settings should not be selected at the same box. it should be possible for both to be visible at the same time, or both to be collapsed to allow more space for the editor. We should be able to select edges of the tables as well. We should be able to drag the borders of the table headers, with a checkbox to use it to set the columns: (x%,y%,z%,...,auto) ratio (default to auto). The cells should have code highlighting. The background/alignment/padding should be precise optional args that can be passed to `#table`. The UI should also be as compact as possible. And we need to detect and parse arbitrary tables. If there is a `#table` then we must do a correct reading into our editor. We need to also have an equivalent tex table editor, but the options will vary depending on the package used so this should be a similar but different UI. For this we will need a bunch of fixtures for testing tables in tex and typ in manual-tests/. Split this todo into individual shippable todo steps before working on it - write it below this one.

Notation	**Meaning and reference**
$m(r) = r^{2\alpha}\ell(r)$	Admissible diffusion, $\frac{1}{2} < \alpha \le \frac{5}{4}$; the additional condition $\ell \in L^2\big(\mathrm{d}\frac{r}{r}\big)$ is imposed only at $\alpha = \frac{5}{4}$. See def:admissible.
H^s_σ, $\mathcal{V}_m$, $\mathcal{V}'_m$	Divergence-free Sobolev space, energy space, and its dual; see the definitions preceding def:leray-hopf.
$\mathbb{P}$, $\hat{v}$, ζ	Leray projection and the Fourier transform with frequency variable ζ; convention after eq:main.
$R(t)$, R_τ, τ, ξ	$m(R(t)) = t^{-1}$, $R_\tau = R(e^\tau)$, $\tau = \log t$, and $\xi = R(t)x$; see eq:ansatz.
M_τ, $b(\tau)$	Normalised diffusion $M_\tau(\rho) = \frac{m(R_\tau \rho)}{m(R_\tau)}$ and scale speed $b = -\partial_\tau \log R_\tau$; see eq:Mtau-b-def.
$\overline{\Omega}$, $\overline{U}$, β	Fixed unstable Euler vorticity, its Biot–Savart velocity, and the large amplitude in prop:frozen-dichotomy.
$\boldsymbol{K}$, $\boldsymbol{T}_{\alpha,\beta}$	Linearised transport and frozen similarity generator in eq:frozen-operators.
$\boldsymbol{P}_{\alpha,\beta}$, $E_{\alpha,\beta}$, $F_{\alpha,\beta}$	Real finite-rank unstable spectral projection, its range, and its kernel in prop:frozen-dichotomy.
U^L, γ, τ_*	Ancient linear mode, decay rate $\gamma > \frac{1}{2}$, and terminal logarithmic time; all conclusions are in prop:ancient-unstable-mode.

$U^{(\pm)}$, F_β	The two profiles $\beta\overline{U} \pm U^L$ and the background profile force in eq:golovkin-profiles.
$u^{(\pm)}$, f, t_*	Physical fields in eq:physical-golovkin-pair, with $t_* = e^{\tau_*}$.

24. [X] improve typst table editor: this isn't detected as a table for some reason

```
#table(
columns: (24%, 76%),
align: left,
inset: (x: 4pt, y: 5pt),
stroke: none,
table.hline(stroke: 0.8pt),
[#strong[Notation]], [#strong[Meaning and reference]],
table.hline(stroke: 0.5pt),
[$m(r)=r^(2alpha) ell(r)$],
[Admissible diffusion, $1/2<alpha<=5/4$; the additional condition
  $ell in L^2 (dif r/r)$ is imposed only at $alpha=5/4$.
  See @def:admissible.],
[$H_sigma^s$, $cal(V)_m$, $cal(V)'_m$],
[Divergence-free Sobolev space, energy space, and its dual;
  see the definitions preceding @def:leray-hopf.],
[$PP$, $hat(v)$, $zeta$],
[Leray projection and the Fourier transform with frequency variable
  $zeta$; convention after @eq:main.],
[$R(t)$, $R_tau$, $tau$, $xi$],
[$m(R(t))=t^(-1)$, $R_tau=R(e^tau)$, $tau=log t$, and $xi=R(t)x$;
  see @eq:ansatz.],
[$M_tau$, $b(tau)$],
[Normalised diffusion $M_tau (rho)=m(R_tau rho)/m(R_tau)$ and
  scale speed $b=-partial_tau log R_tau$; see @eq:Mtau-b-def.],
[$overline(Omega)$, $overline(U)$, $beta$],
[Fixed unstable Euler vorticity, its Biot--Savart velocity, and the
  large amplitude in @prop:frozen-dichotomy.],
[$bold(K)$, $bold(T)_(alpha,beta)$],
[Linearised transport and frozen similarity generator in
  @eq:frozen-operators.],
[$bold(P)_(alpha,beta)$, $E_(alpha,beta)$, $F_(alpha,beta)$],
[Real finite-rank unstable spectral projection, its range, and its
  kernel in @prop:frozen-dichotomy.],
[$U^L$, $gamma$, $tau_*$],
[Ancient linear mode, decay rate $gamma>1/2$, and terminal logarithmic
  time; all conclusions are in @prop:ancient-unstable-mode.],
[$U^((plus.minus))$, $F_beta$],
[The two profiles $beta overline(U) plus.minus U^L$ and the background
  profile force in @eq:golovkin-profiles.],
[$u^((plus.minus))$, $f$, $t_*$],
[Physical fields in @eq:physical-golovkin-pair, with $t_*=e^(tau_*)$.],
table.hline(stroke: 0.8pt),
)
```

25. [X] improved git hunk popup: turn into a better editor:
    1. allow lines to be individually reverted in a hunk
    2. allow manual editing inside the hunk editor
    3. allow dragging around / resizing the hunk
26. [X] allow right-click on math brings up right-click menu with `> convert to... >` with the options of inline math, display math. In TeX, can choose between inline math (`\(\)`) and the various display envs (`align, align*, gather, gather*, equation, equation*`)

27. [X] improve the terminal panel - why cant i interact with tui apps like opencode properly?
28. [X] terminal panel needs a button to open more terminals
29. [X] when a draggable divider is highlighted, the handle needs to be outlined in the same color as well
30. ☐ impl translucency and blur of windows with gpui. maybe pull in a dependency
31. ☐ compile a version of our app that also works for intel macs, make a new script for this and put it in scripts/ and test the compilation
32. ☐ improvement to right-click: when text is highlighted, right-click should not deselect the text. In particular we should be able to right-click and copy text
33. ☐ right click on typst math should not have a convert to... option as there is only one choice - if it is inline math, then it should simply be convert to display math. if it is display math, it should simply be convert to inline math.
35. ☐ system theme mode (dark/light) (+ theme colors if it makes sense) should be propogated to the termina panel. opencode does not currently detect the right theme.
36. ☐ optional dark mode for previewed pdfs and images
37. ☐ bug in the draggable divider - in extreme shapes the